National Research University Higher School of Economics (HSE)

Curriculum

Field of study 01.03.02 Applied Mathematics and Information Science

Educational Programme "Data Science and Business Analytics"

Specializations: "Data Analysis in Applied Research", "Data Science in Business", "Data Science in Finance"

Implementing unit: Faculty of Computer Science, HSE - Moscow

2 nd, 2025/2026 academic year

APPROVED

06.06.2025

Vice Rector

ROSHCHIN S.Y.

Signed with EDS

Length of Programme: 4 years

Years of Study: 2024/2025 - 2027/2028

Mode of Study: Full Time

Degree: Bachelor's degree

Block Code	Course	Subject type	Department	Period of execution	Credits	Total Academic Hours	Contact Hours	Allocation of Contact Hours				Additional Information
								1	2	3	4	
	Degree Progamme				**60,00**	**2 442**	**1 130**	**224**	**238**	**424**	**244**	
	Data Culture						**2**				**2**	
1	Independent Programming Test. Advanced Level	C	отдел развития цифровых компетенций				2				2A	Online Course, Foreign language
	Major				**41,00**	**1 558**	**858**	**168**	**182**	**340**	**168**	
	Core Professional Block (Major)				**26,00**	**988**	**598**	**140**	**154**	**220**	**84**	
	Core Professional Block (Major)				**26,00**	**988**	**598**	**140**	**154**	**220**	**84**	
1	Algorithms and Data Structures	C	Department of Big Data and Information Retrieval		6,00	228	192	56A	56	80A		Foreign language
2	Computer Architecture and Operating Systems	C	Department of Big Data and Information Retrieval		4,00	152	136			80	56A	Foreign language
3	Discrete Mathematics 2	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language
4	Mathematical Statistics	C	Department of Big Data and Information Retrieval		4,00	152	88			60	28A	Online Course, Foreign language
5	Calculus 2	C	Department of Big Data and Information Retrieval		4,00	152	56	28	28A			Foreign language
6	Probability Theory	C	Department of Big Data and Information Retrieval		4,00	152	70	28	42A			Online Course, Foreign language
	Elective Professional Block (Major)				**15,00**	**570**	**260**	**28**	**28**	**120**	**84**	
	Compulsory Courses				**15,00**	**570**	**260**	**28**	**28**	**120**	**84**	
1	Differential Equations	C	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
2	Machine Learning 1	C	Department of Big Data and Information Retrieval		4,00	152	68			40	28A	Foreign language
3	Introduction to Micro and Macroeconomics	C	Department of Big Data and Information Retrieval		7,00	266	124	28	28A	40	28A	Foreign language
	Minor				**10,00**	**380**	**124**	**28**	**28**	**40**	**28**	
1	Minor	E			10,00	380	124	28	28A	40	28A	Foreign language
	General Courses				**1,00**	**200**	**144**	**28**	**28**	**44**	**44**	

	Core General Courses				**1,00**	**200**	**144**	**28**	**28**	**44**	**44**	
1	Physical Training	C	Department of Physical Training		1,00	200	144	28	28	44	44A	Foreign language
	Internship				**8,00**	**304**	**2**				**2**	
	Project Internship				**5,00**	**190**	**2**				**2**	
1	Programming Project 1	C		F	5,00	190	2				2A	Foreign language
	Professional Internship				**3,00**	**114**						
1	Educational Internship	C		F	3,00	114						Foreign language

Curriculum agreed:

Academic Supervisor	Voznesenskaia T.V.	23.05.2025
Dean	ARZHANTSEV I.V.	23.05.2025
Head of Centre for Educational Model Design	LEPESHKIN I.A.	06.06.2025

* Subject type:

Compulsory course	C
Elective course	E

Period of execution:

Fixed	F